

Inverted perovskite solar cells using dimethylacridine-based dopants

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Doping of perovskite semiconductors¹ and passivation of their grain boundaries² remain challenging but essential for advancing high-efficiency perovskite solar cells. Particularly, it is crucial to build perovskite/indium tin oxide (ITO) Schottky contact based inverted devices without predepositing a layer of hole-transport material^{3–5}. Here we report a dimethylacridine-based molecular doping process used to construct a well-matched *p*-perovskite/ITO contact, along with all-round passivation of grain boundaries, achieving a certified power conversion efficiency (PCE) of 25.39%. The molecules are shown to be extruded from the precursor solution to the grain boundaries and the bottom of the film surface in the chlorobenzene-quenched crystallization process, which we call a molecule-extrusion process. The core coordination complex between the deprotonated phosphonic acid group of the molecule and lead polyiodide of perovskite is responsible for both mechanical absorption and electronic charge transfer, and leads to *p*-type doping of the perovskite film. We created an efficient device with a PCE of 25.86% (reverse scan) and that maintained 96.6% of initial PCE after 1,000 h of light soaking.

Perovskite solar cells (PSCs) have undergone considerable advancement over the past decade, with a certified power conversion efficiency (PCE) of over 25% (refs. 6–8), and can be classified into two types—normal and inverted. With regard to investment, the choice of technical direction is predominant. High-efficiency normal devices suffer from the inevitable degradation of the indispensable and doped Spiro-OMeTAD hole-transport layer (HTL)⁹. By contrast, inverted devices have a simpler device structure and greater HTL material options, both organic and inorganic. However, regarding HTL, popular polymers such as poly(3,4-ethylenedioxythiophene)-poly(styrenesulfonate) and poly(triary amine) also suffer from light- and heat-induced degradation, along with moisture damage^{10,11}. Popular inorganic NiO HTL materials will react with perovskite and cause interfacial decomposition¹². To address the above concerns, the recent advancement of inverted PSCs without predepositing HTL^{3–5,13} avoids degradation of HTL materials and interfacial decomposition, and reduces the cost of materials and related manufacturing processes. In that device architecture, in principle the perovskite film should be *p*-type and construct a *pn* junction with an electron-transport layer, such as C₆₀ and its derivatives, to separate electron–hole (*e–h*) pairs. *p*-dopants such as tetracyanoquinone³, CuSCN¹³ and Cu(thiourea)I² have been employed to modulate *p*-type perovskite films and facilitate effective hole extraction based on band bending at the perovskite/indium tin oxide (ITO) interface^{3,4}. However, carrier transport behaviour at these perovskite/ITO Schottky contacts and the spatial distribution of *p*-dopants were hardly investigated, and their device performances are inferior to that of routine inverted PSCs⁷. In addition, unpassivated perovskite grain boundaries are full of deep-level traps^{14,15}. Therefore, one requires an understanding of

the working mechanisms of *p*-dopant molecules in terms of perovskite film formation on ITO and energy-level alignment of the device. Only on that basis can we design the molecular structure of *p*-dopants to achieve advances in device performance.

Design of the DMacPA molecule

Learning from the features of carbazole phosphonic acid molecules^{16–18}, the phosphonic acid group is projected to form a strong Pb–O bond in coordination while the conjugated group with a suitable energy level is kept as acridine (Ac) in the designed molecule, along with two C–Br groups. The chemical structure of (4-(2,7-dibromo-9,9-dimethylacridin-10(9H)-yl)butyl)phosphonic acid (DMacPA) is shown in Fig. 1a and Supplementary Fig. 1. The two methyl groups of DMacPA create a steric effect that prevents aggregation and jamming at grain boundaries in the film-formation process and yielding the recombination centres. As predicted by density functional theory (DFT) calculation on π – π stacking of molecules (Supplementary Fig. 2), in a dimer model the AcPA molecule with dihydrogen (DHAcPA) tends to stack readily and aggregate. With the dimethyl group (Supplementary Fig. 2j), the required binding energy of the DMacPA dimer is 31 meV higher than that of DHAcPA, along with a greater distance of 4.01 Å, which suggests that DMacPA is unlikely to form dimers and aggregate further. Because the greater intermolecular distance (around 4.22 Å) in the diethyl analogue (DEAcPA) may render electron hopping more difficult, DMacPA was therefore selected. The DMacPA structure was determined by nuclear magnetic resonance (NMR) (Supplementary Figs. 3–9). DMacPA shows a robust decomposition temperature of

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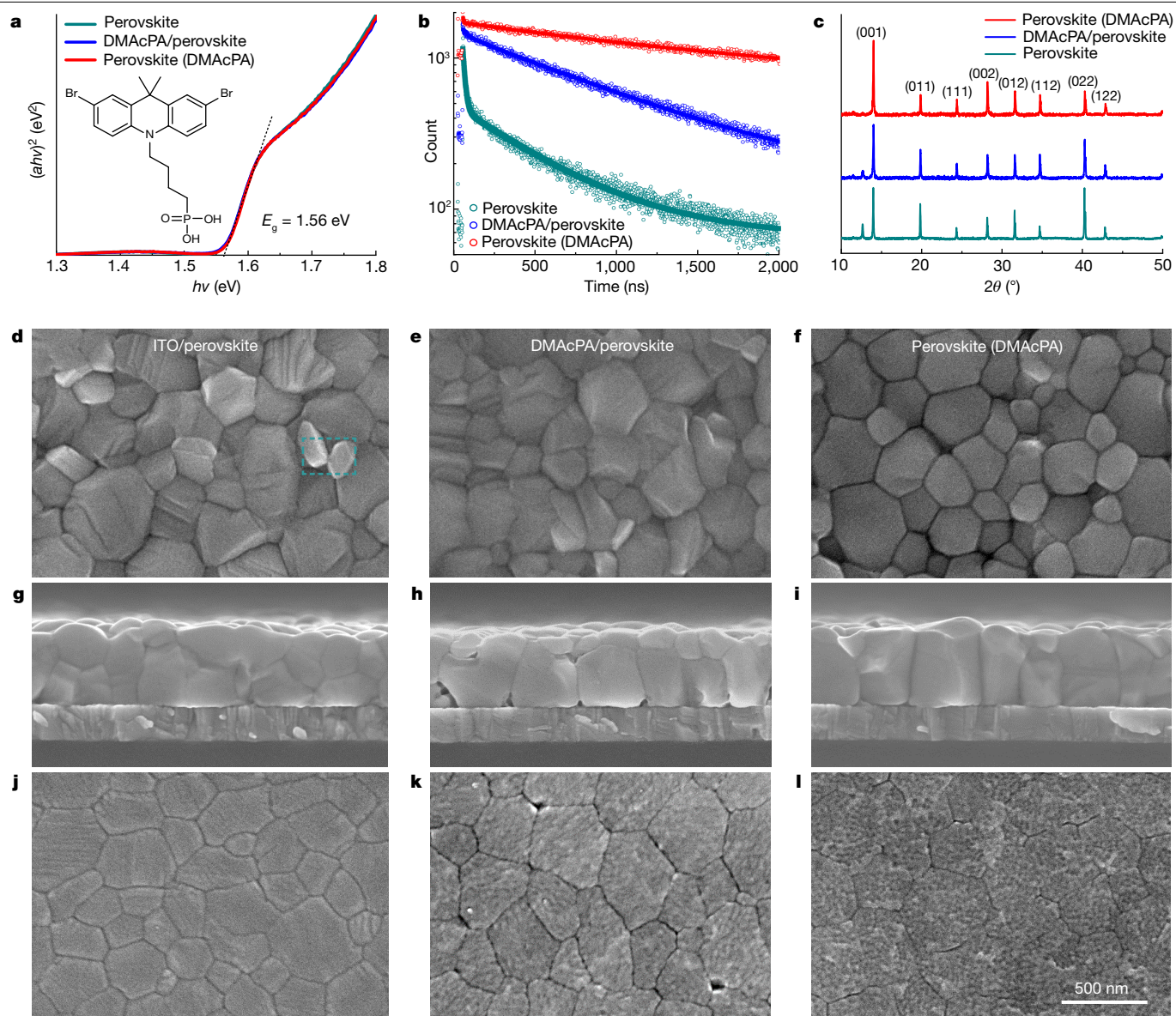


Fig. 1 | Structure and property characterization of three perovskite films. **a**, Tauc plots of absorption spectra of three types of film deposited on ITO substrates, with the chemical structure of DMACPA inset. **b**, Time-resolved photoluminescence decay spectra of films deposited on glass. **c**, X-ray

diffraction patterns of perovskite films deposited on ITO substrates. **d–l**, Top view (**d–f**), cross-sectional (**g–i**) and bottom view (**j–l**) scanning electron microscopy (SEM) images of films from the ITO/perovskite (**d,g,j**), DMACPA/perovskite (**e,h,k**) and perovskite (DMACPA) (**f,i,l**) samples.

217.8 °C (Supplementary Fig. 10) and an optical band gap of 3.65 eV (Supplementary Fig. 11). The lowest unoccupied molecular orbital (LUMO) and highest occupied molecular orbital (HOMO) are mainly located on the conjugated Ac group (Supplementary Fig. 12). Using ultraviolet photoelectron spectroscopy (UPS) (Supplementary Fig. 13), its work function (WF), HOMO and LUMO were determined at –5.48, –5.98 and –2.33 eV, respectively, which makes it a suitable hole-extraction material for PSCs^{16–18}. The above characteristics render DMACPA a potential dopant for realization of our molecule-extrusion process.

Molecule-extrusion process

To examine the molecule-extrusion process we prepared Cs⁺, formamidinium (FA⁺) and methylammonium (MA⁺) mixed-cation perovskite, with an optical band gap of 1.56 eV, directly on an ITO substrate (Supplementary Fig. 14 and Fig. 1a)¹⁹, including a DMACPA-doped perovskite

film directly deposited on ITO (perovskite (DMACPA)) as the target, both undoped pristine perovskite films deposited directly on bare ITO (perovskite) and on a DMACPA layer-modified ITO (DMACPA/perovskite) as references. Besides enhanced photoluminescence intensity (Supplementary Fig. 15), the doped perovskite film had the longest lifetime (1,926 ns) whereas that of the pristine perovskite was 498 ns (Fig. 1b and Supplementary Table 1). The augmented peak-intensity ratio of (001)/(022) indicates that DMACPA doping may induce some oriented growth of perovskite (Fig. 1c). A more obvious change is the attenuated peak intensity of excess PbI₂ (12.5°). Along with the sharp drop in PbI₂ peak intensity with varying DMACPA dosage (Supplementary Fig. 16), they both demonstrate that DMACPA coordination can convert crystalline PbI₂ to amorphous PbI₂-DMACPA chelates⁶. Since perovskite crystallization occurs from top to bottom in the chlorobenzene-quenching process²⁰, we can screen the top and buried bottom surfaces of perovskite to reveal the traces of DMACPA and its coordinated PbI₂.

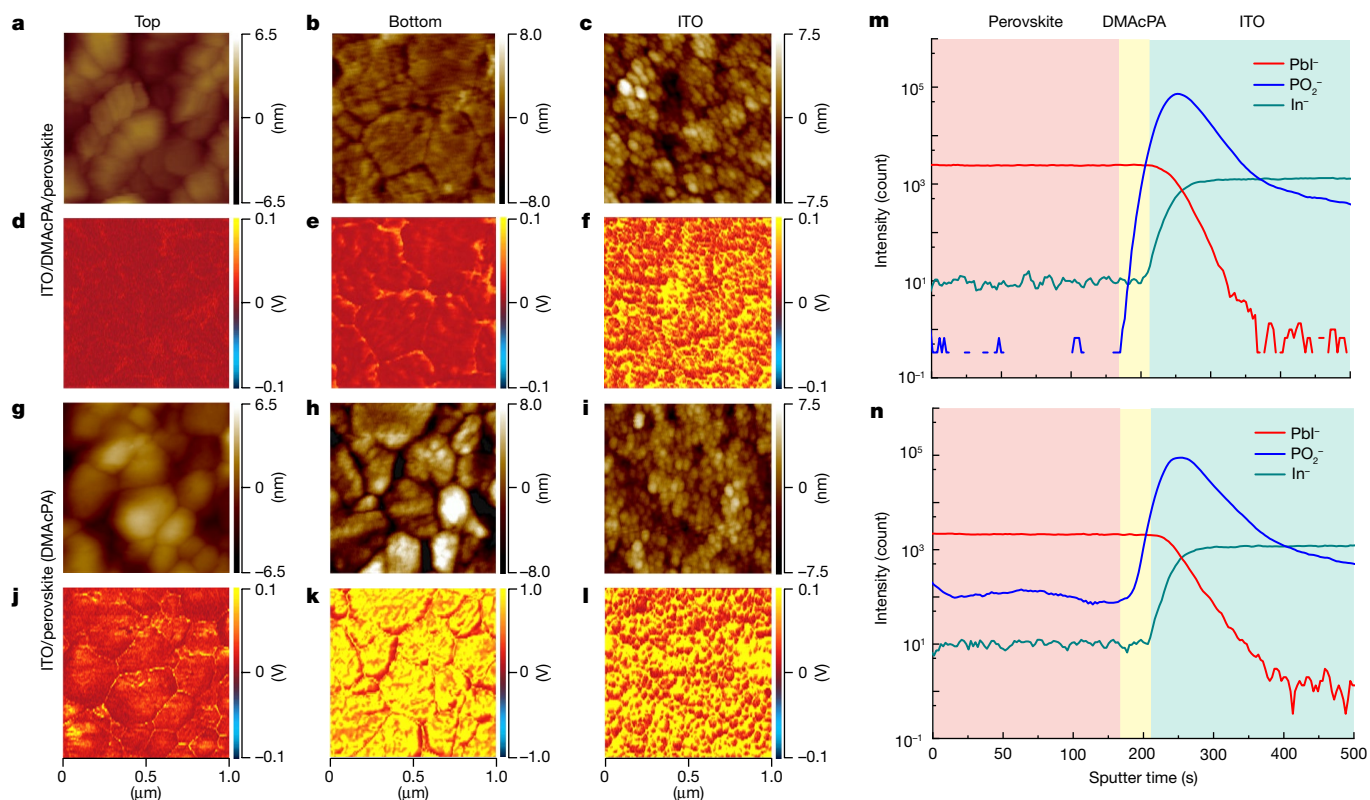


Fig. 2 | AFM-IR and ToF-SIMS provide evidence for the molecule-extrusion mechanism. **a–f**, AFM-IR topographic and corresponding $1,415\text{ cm}^{-1}$ absorption images of top surface (**a,d**), bottom surface (**b,e**) and detached ITO surface (**c,f**) of the ITO/DMAcPA/perovskite (undoped) film. **g–l**, AFM-IR topographic

and corresponding $1,415\text{ cm}^{-1}$ absorption images of top surface (**g,j**), bottom surface (**h,k**) and detached ITO surface (**i,l**) of the ITO/perovskite (DMAcPA) film. **m,n**, ToF-SIMS profiles of ITO/DMAcPA/perovskite (**m**) and ITO/perovskite (DMAcPA) (**n**) samples.

Depending on the peel-off process by a solidified glue (Supplementary Fig. 17)²², Fig. 1d–f shows the smoother morphology and diminished excess PbI_2 flakes (highlighted in a green frame in Fig. 1d and Supplementary Fig. 18), identified by energy-dispersive X-ray spectroscopy results (Supplementary Tables 2 and 3). Compared with Fig. 1g,i, Fig. 1h shows that some voids exist at the interface between the undoped perovskite and DMAcPA-modified ITO, due to the more hydrophobic surface of the DMAcPA layer (Supplementary Fig. 19). This is also verified by the buried bottom surfaces shown in Fig. 1j–l. The most important information derived from the buried bottom surfaces is the crumpled morphology of DMAcPA/perovskite (Fig. 1k) and perovskite (DMAcPA) (Fig. 1l), in sharp contrast to the clean bottom surface of the perovskite (Fig. 1j). Correlating the fall-off in excess PbI_2 (Fig. 1c), we speculate that the dopant molecules may be extruded to the bottom surface of the film in the chlorobenzene-quenching process.

Atomic force microscopy-based infrared spectroscopy (AFM-IR) was employed to probe the critical bottom and top surfaces of films (Fig. 2a–l), depending on the feature peak of P–C stretching vibration of DMAcPA at $1,415\text{ cm}^{-1}$ (Supplementary Fig. 20). The detached ITO surface of perovskite (DMAcPA) (Fig. 2l) shows a higher amount of DMAcPA molecules than in DMAcPA/perovskite (Fig. 2f), suggesting that chelates of DMAcPA– PbI_2 are extruded to the bottom surface, combining the appearance of Pb, I and P elements at the detached ITO surface X-ray photoelectron spectroscopy (XPS) signal (Supplementary Fig. 21) for the doped perovskite. The molecule contrast between the two bottom surfaces (Fig. 2e,k) is so sharp that we had to scale up the vertical scale bar tenfold for perovskite (DMAcPA). That demonstrates that DMAcPA molecules adhere to the perovskite grain surface in the extrusion process. When we increased doping concentration from 4 to 10 mg ml^{-1} , grain boundaries were expanded with gaps at both the top

and bottom surface (Supplementary Fig. 22), showing that DMAcPA exists also at grain boundaries throughout the doped perovskite film. This was also verified by the characterization results of time-of-flight secondary ions mass spectroscopy (ToF-SIMS) (Fig. 2m,n). Thus far, our work clearly shows the spatial distribution of p-dopant molecules in the doped perovskite film, as rarely discussed in previous reports of so-called HTL-free PSCs^{3–5,13}.

Because all the above findings indicate the critical coordination between DMAcPA and perovskite, we need to investigate the mode of coordination. pH testing is a useful method to show the concentration of solvated protons²². Supplementary Table 4 unambiguously shows that deprotonation of DMAcPA occurs in its dissolution process. In the perovskite solution, solvated $[\text{H}^+]$ increased by over three orders of magnitude after dissolving DMAcPA, with pH decreasing from 7.57 to 4.12. Moreover, both PbI_2 and FAI can stimulate the ionization level of DMAcPA. The corresponding ^1H NMR results confirm the ionization of DMAcPA in solution (Fig. 3a), in which active hydrogen moved the downfield chemical shift from 2.65 to 4.88 ppm when the solvent was changed from CDCl_3 (poor solvent) to DMF-d_7 (good solvent). In concert with the above pH results, it is shown that the protons that escaped from the phosphonic acid group were probably captured by DMF due to the formation of solvated $\text{H}^+[\text{DMF}]_x$. With the incorporation of PbI_2 and FAI the peak changed moderately to an upfield chemical shift, which demonstrates that I^- ions are inclined to form the $\text{H}^+[\text{DMF}]_x\text{I}^-$ complex²³. A higher concentration of iodine input led to a bigger shift (Supplementary Fig. 23), consistent with the above pH data and in agreement with the upfield shift of the ^{31}P resonance signal (Supplementary Fig. 24)²⁴.

To explore the coordination between DMAcPA and perovskite we carried out first-principles calculations considering the spatial effect of

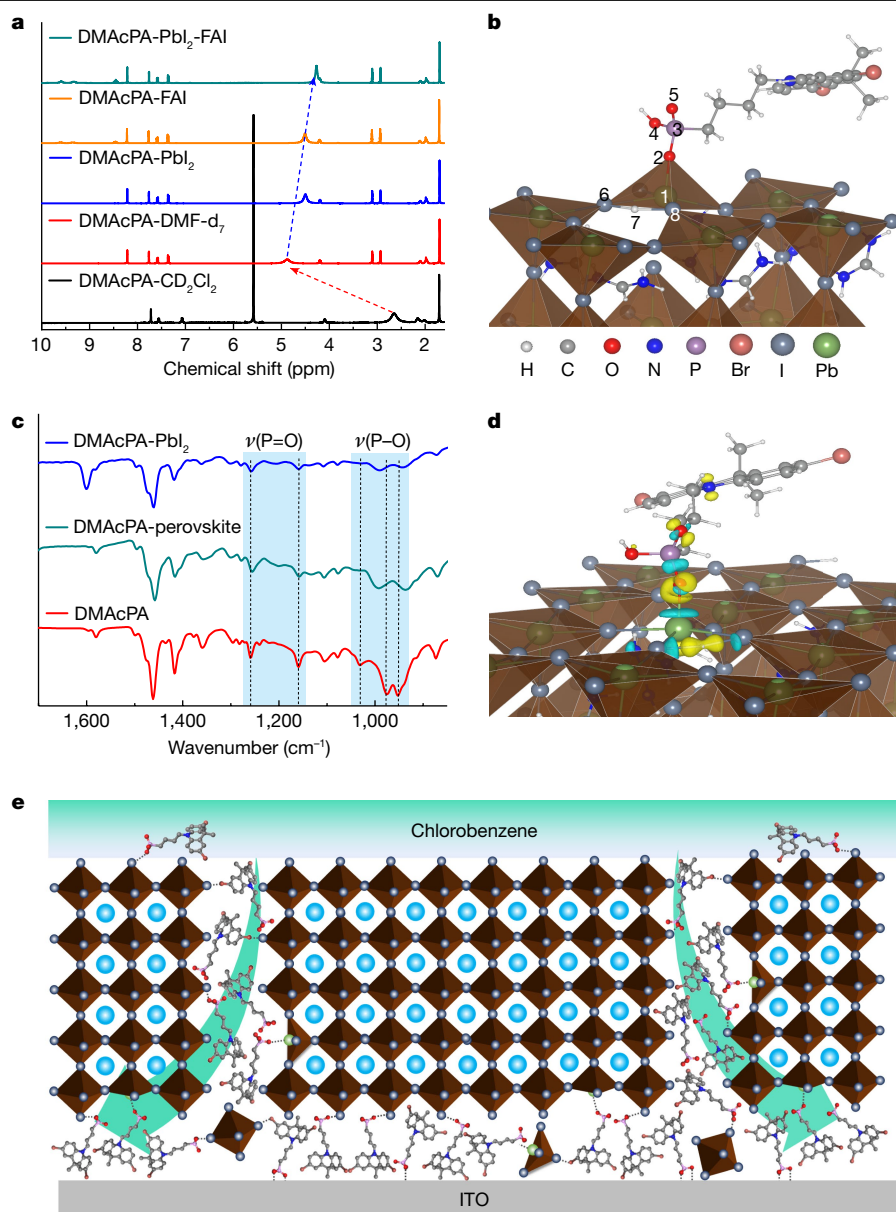


Fig. 3 | Investigation of molecule-extrusion mechanisms. a, ^1H NMR spectra of DMAcPA in CD_2Cl_2 and DMF-d_7 with different iodides at the same molar dosage. **b,** Adsorption structure of deprotonated DMAcPA absorbed on the PbI_2 -terminated FAPbI_3 crystal surface. **c,** ATR-IR spectra of pure DMAcPA and

when combined with either PbI_2 or perovskite. **d,** DFT-predicted electron transfer between DMAcPA and FAPbI_3 . Yellow and cyan denote electron accumulation and depletion, respectively. **e,** Schematic illustration of the molecule-extrusion model.

solvated cations^{25,26}—for example, $\text{H}^+[\text{DMF}]_x$ and $\text{H}^+[\text{DMSO}]_x$. Figure 3b shows that the stable adsorption configuration exhibits a coordination between ionized $-\text{P}-\text{O}^-$ and Pb, with a bond length (d_{1-2}) of 2.18 Å. The $-\text{P}-\text{O}^-$ covalent bond (d_{2-3}) involved was consequently shortened from 1.63 to 1.57 Å and the $-\text{P}-\text{O}(\text{H})$ bond (d_{3-4}) was elongated from 1.62 to 1.64 Å (Supplementary Table 5). The bond length of $-\text{P}=\text{O}$ (d_{3-5}) remained almost unchanged (1.49 Å). These structural changes are in good agreement with our FTIR characterizations (Fig. 3c). Specifically, the blue (976 cm^{-1})- and red (956 cm^{-1})-shifted peaks of P–O bonds correspond to shortened P–O[−] and elongated P–O(H) bonds, respectively. The infrared peak of the P=O bond ($1,159$ and $1,260\text{ cm}^{-1}$) exhibits a negligible shift (Fig. 3c), consistent with the unchanged P=O bond. For comparison we also examined the adsorption of neutral DMAcPA on the perovskite surface (Supplementary Fig. 25). In this case the coordination of $-\text{P}=\text{O}$ with Pb was the most stable configuration, in which the P=O bond was elongated by about 0.02 Å and the two P–O(H)

bonds both shortened by roughly 0.02 Å (Supplementary Fig. 25 and Supplementary Table 5). These results are not consistent with the negligible shift of the P=O peak and the red- and blue-shifted P–O peaks observed in the infrared spectrum. Therefore, both our DFT calculations and FTIR results demonstrate that adsorption of DMAcPA on the PbI_2 -terminated perovskite crystal surface occurred through bonding between $-\text{P}-\text{O}^-$ and Pb. This is also indicated by the red shift of Pb²⁷ (Supplementary Fig. 26a) and I (Supplementary Fig. 26b) in the binding energy detected at the buried bottom surface.

Based on the foregoing, we present the molecule-extrusion mechanism in Fig. 3e. In the chlorobenzene-quenching process, the dative bonds of Pb–O between DMAcPA and perovskite are dissociated by the stronger Pb–I bonding of $[\text{PbI}_6]^{4-}$ cages, and molecules are extruded to the crystal surface because their large size and negative polarity prevent DMAcPA from entering the perovskite lattice^{28,29}. Because the perovskite film crystallizes from top to bottom²⁰, DMAcPA and

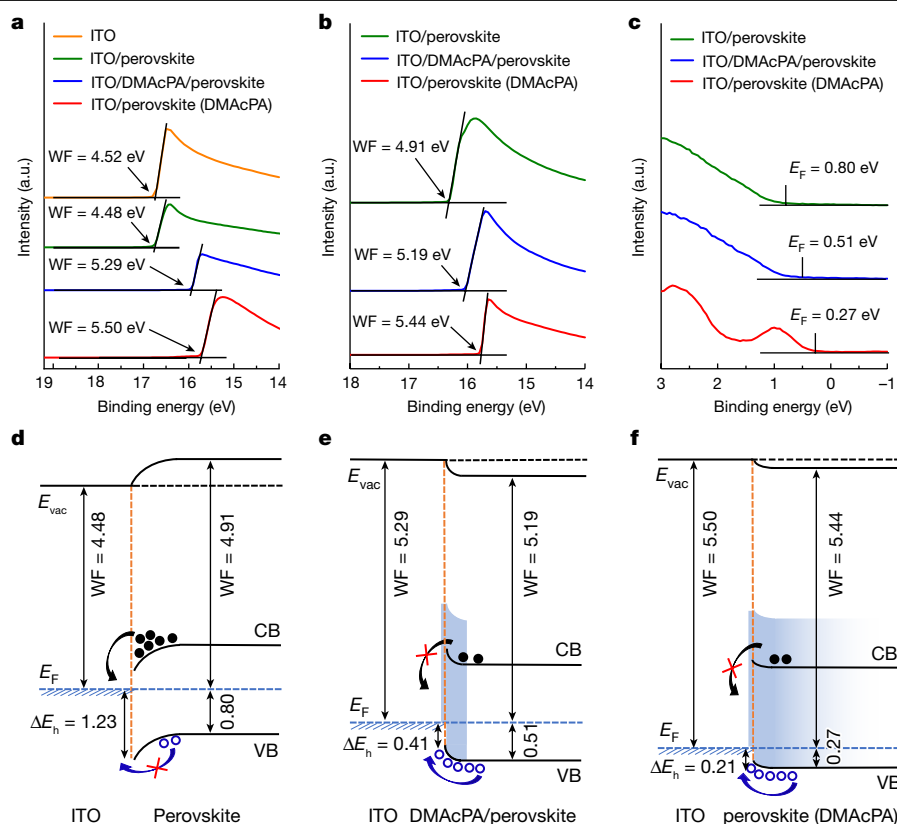


Fig. 4 | Investigation of the hole-transfer mechanism at perovskite-ITO Schottky contacts. a–c, Cutoff region of UPS for bare ITO and detached ITO surfaces (a), top surfaces (b) and onset region of top surfaces (c) of the three

types of perovskite film. **d–f,** Energy-level diagrams for ITO/perovskite (d), ITO/DMAcPA/perovskite (e) and ITO/perovskite (DMAcPA) (f). a.u., arbitrary units, WF, work function; E_F , Fermi level; CB, conduction band; VB, valence band.

its chelates with PbI_x are extruded to the bottom surface of the film. The unique autogenic layer can be viewed as an indivisible part of the perovskite absorber film, tailoring the interface between perovskite and ITO both chemically and electronically.

***p*-Perovskite/ITO Schottky contact**

We then utilized UPS to define the conduction type of perovskite and investigate band bending at the interface between perovskite and ITO (Fig. 4a–c), which is claimed to be responsible for hole extraction^{3,4}. Compared with -4.52 eV of our bare ITO, the surface work function of the pristine perovskite-contaminated ITO moved upwards marginally to -4.48 eV whereas those of DMAcPA/perovskite and perovskite (DMAcPA) moved downwards to -5.29 and -5.50 eV, respectively. It is clear that DMAcPA and its chelates with PbI_x modify the surface work functions of ITO substrates^{3,16–18,30}. The undoped perovskite deposited on the *p*-type DMAcPA layer showed a weak *p* type (Fig. 4e) whereas the pristine perovskite on the bare ITO tended to be neutral (Fig. 4d). The doped perovskite showed a strong *p* type (Fig. 4f), which can be attributed to the charge-transfer complex mechanism of the molecular doping mechanism^{30–32}. The charge-transfer complex in this work is the coordinating complex of the phosphonic acid group and Pb(I)_x polyhedron. The charge density difference illustrated in Fig. 3d further shows that electrons accumulate around the hydrogen proton from DMAcPA and that oxygen bonded with the perovskite. Bader charge analysis indicated that 0.7 electrons were transferred from perovskite to DMAcPA. Decreased electron density (n) must increase hole density (p), subject to a constant product (np) following thermal equilibrium at one temperature³³. This resulted in the *p*-type character of our DMAcPA-doped perovskite films as examined by a positive Hall coefficient tested.

According to the band alignment of Schottky contact³³, for ITO/perovskite (Fig. 4d) the perovskite band bends downward and creates a barrier to hole extraction but an offset favouring electron transfer to ITO, resulting in a low device PCE of 7.75% (Supplementary Fig. 27). For ITO/DMAcPA/perovskite (Fig. 4e), the perovskite band bends upwards but with a large offset (ΔE_h) of 0.41 eV. Band bending can facilitate hole extraction and block electrons whereas a high ΔE_h would undermine hole transfer rate due to decreased interfacial concentration and open-circuit voltage by thermionic loss^{30,33}. For ITO/perovskite (DMAcPA) (Fig. 4f), the perovskite band shows an upward bending along with a much lower ΔE_h of 0.21 eV owing to its stronger *p* type and deeper Fermi level, which boosts its hole-extraction rate at the interface.

Device performance and analysis

As shown in Fig. 5a, the target device achieved a PCE of 25.86% in reverse scan with 1.187 V in open-circuit voltage (V_{oc}), 25.69 mA cm^{-2} in short-circuit current density (J_{sc}) and 84.73% in fill factor (FF). By contrast, following optimization (Supplementary Fig. 28) the control device had a PCE of 23.03% in reverse scan with 1.132 V in V_{oc} , 25.45 mA cm^{-2} in J_{sc} and 79.87% in FF. The J_{sc} values agree with those integrated from external quantum efficiency (EQE) spectra (Fig. 5b), with a small deficiency gap³⁴. These 500-s-scale, steady-state PCEs prove the reliability of our test (Supplementary Fig. 29). Data from a batch of 24 devices for each type show an average PCE of 25.45 and 22.50% for target and control, respectively (Supplementary Fig. 30 and Supplementary Table 6). The certified device PCE by the third party (Shanghai Institute of Microsystem and Information Technology) was 25.39% (Supplementary Fig. 31). To examine the steric effect, a

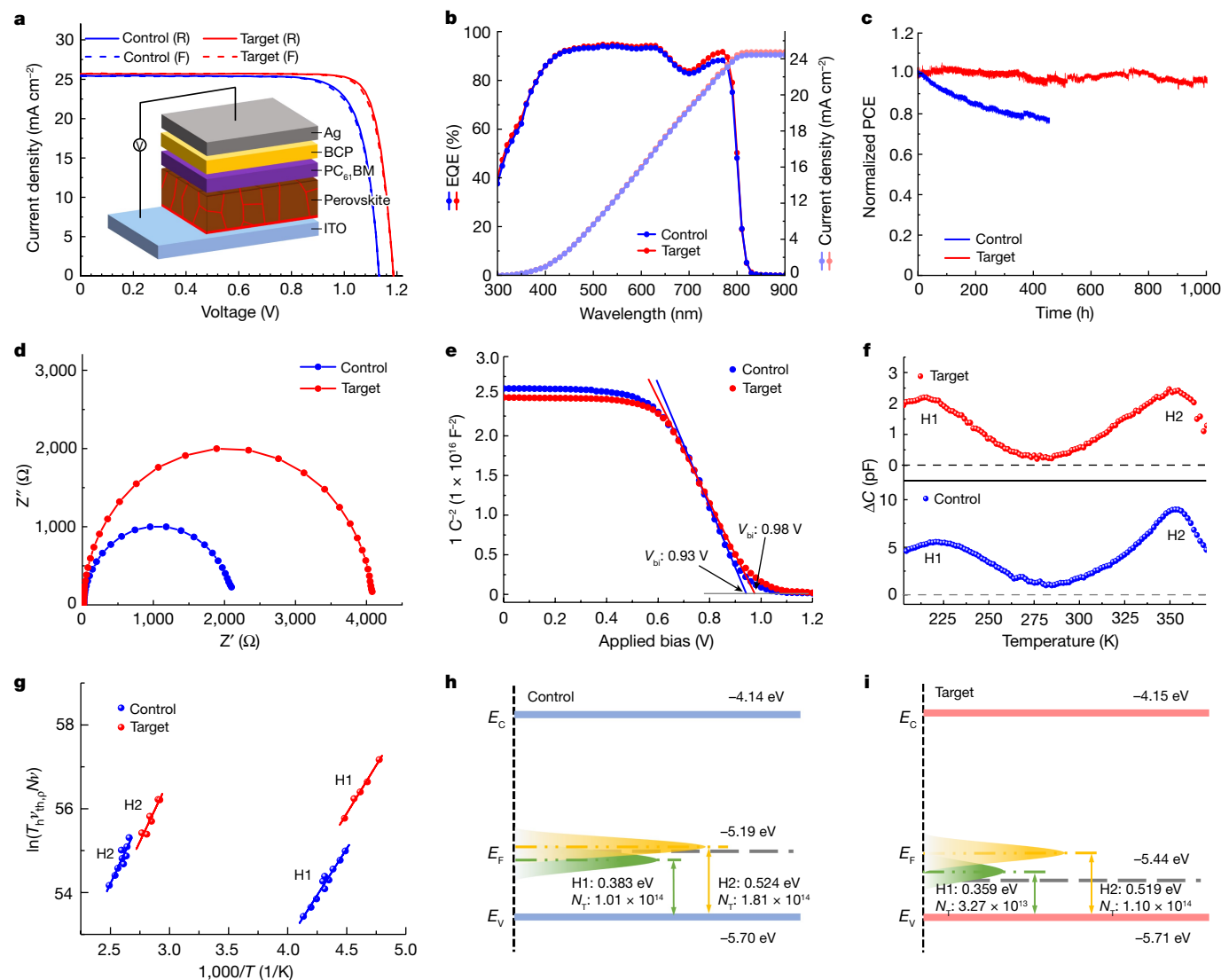


Fig. 5 | Device performance tests and analysis of PSCs based on ITO/DMAcPA/perovskite (control) and ITO/perovskite (DMAcPA) (target) Schottky contacts. a, J – V characteristics of best-performing PSCs; the inverted device structure was inserted. b, EQE spectra. c, Continuous maximum-power point tracking for encapsulated PSCs under AM 1.5 illumination in ambient air. d, e, Nyquist (d) and Mott–Schottky (e) plots. f, g, Deep-level transient

spectroscopy (DLTS, f) and corresponding Arrhenius plot (g) with fitting. h, i, Energy-level diagrams of band (h) and defect (i) of corresponding devices. Corresponding E_c , E_v and E_f data are derived from Fig. 4e, f. Z' , real part of the impedance; Z'' , imaginary part of the impedance; N_D , concentration of deep-level traps measured.

MeO-2PACz dimer with minimal layer distance of about 3.53 Å and low required binding energy of 14 meV (Supplementary Fig. 2f), and a bigger inclination to aggregate, resulted in a target device performance of only 19.46% while its control device achieved a reasonable 22.26% (Supplementary Figs. 32 and 33)⁷. In the branch of PSCs without predepositing HTLs, our molecule-extrusion process raised device performance to a new level (Supplementary Fig. 34). Our champion device came from the optimized dosage of DMAcPA (Supplementary Fig. 35), which was determined by the highest FF at 4 mg ml^{-1} , while the process window of V_{oc} and J_{sc} is broad. This can be ascribed to an electrically non-compact extrusion layer below 4 mg ml^{-1} while the thicker layer would abate FF. The enhanced target device performance also came from the improved bulk quality of perovskite films, as indicated by the longer lifetime of excitons derived from photoluminescence (Fig. 1b) and carriers from transient absorption (Supplementary Fig. 36).

Figure 5c shows the maximum-power point tracking of PCE for target and control devices in a continuous light-soaking test from their initial performance (Supplementary Fig. 37). The target device maintained

96.6% of its initial PCE after 1,000 h whereas the control fell to below 80% after 500 h. This excellent long-term stability can be attributed to the all-round passivation of grain boundaries by DMAcPA (Fig. 2 and Supplementary Fig. 22)^{14,15}. This was also demonstrated by the augmented recombination resistance (R_{rec}), from 2,046 to 4,028 Ω , after doping whereas series resistance (R_s) remained almost unchanged (Fig. 5d, Supplementary Fig. 38 and Supplementary Table 7). The built-in field voltages (V_{bi}) deduced from Mott–Schottky plots were 0.98 and 0.93 V for target and control, respectively. This difference is in good agreement with that in V_{oc} derived from the J – V test (Fig. 5a). Irrespective of the target or control, V_{bi} must have arisen more from the perovskite–PC₆₁BM pn junction than from the above band bending. The deeper Fermi level of the doped perovskite film leads to the higher V_{bi} when the work function of PC₆₁BM remained the same³³.

Deep-level trap spectroscopy was applied to measurement of deep-trap energy level and density. Figure 5f shows two positive peaks, at around 220 and 350 K, for the ITO/perovskite (DMAcPA) and ITO/DMAcPA/perovskite devices, which can be determined as hole

traps (H1 and H2) according to the polarity of the bias (−0.3 V) and pulse voltages (+0.4 V) on our p-perovskite-based inverted devices in the measurement system³⁵. According to the Arrhenius equation for hole traps³⁶, we can fit the two spectra and acquire their corresponding Arrhenius plots (Fig. 5g). The activation energies of the H1 and H2 traps of the doped perovskite were then determined, at 0.359 and 0.519 eV above the valence band maximum (Supplementary Table 8), respectively. By contrast, the control perovskite showed deeper H1 (0.383 eV) and H2 (0.524 eV) traps. H1 is related to the interstitials of A-cations whereas H2 can be ascribed to antisite defects of MA_i and Pb_i (refs. 14,15), which occur often at grain boundaries and surfaces. For H1 the target device based on ITO/perovskite (DMAcPA) had a trap density of $3.27 \times 10^{13} \text{ cm}^{-3}$, which is reduced from the $1.01 \times 10^{14} \text{ cm}^{-3}$ of the ITO/DMAcPA/perovskite device. The related capture cross-section also reduced, from $5.93 \times 10^{-16} \text{ cm}^2$ to $7.35 \times 10^{-17} \text{ cm}^2$. For H2 the density also decreased, from $1.81 \times 10^{14} \text{ cm}^{-3}$ to $1.10 \times 10^{14} \text{ cm}^{-3}$, following bulk doping of DMAcPA. The corresponding energy-level diagrams (Fig. 4h,i) show that DMAcPA doping reduced deep-level hole trap density, in particular by suppression of the defect states of grain boundaries and surfaces. Suppression is crucial to enhance the lifetime of waiting holes at the perovskite/ITO interface, due to the undetectably ultralow mobility of the molecule, and also boosts device PCE.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-023-06207-0>.

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Methods

Materials

All reagents and solvents in this work were used as purchased, with detailed information provided in the Supplementary Information. DMAcPA was synthesized in our laboratory, with the detailed synthetic procedure shown in the Supplementary Information.

Device fabrication

The inverted (p-i-n) perovskite solar cell studied in this paper had the architecture ITO/perovskite/[6,6]-Phenyl-C61-butyric acid methyl ester (PC₆₁BM)/Bathocuproine/Ag. The ITO glass substrate was cleaned successively with detergent (2% in deionized water), deionized water, acetone and isopropanol, each for 15 min, in an ultrasonic bath. Next, the substrate was dried with N₂ and treated with ultraviolet ozone for 15 min. For control devices, DMAcPA at a concentration of around 0.5–3.0 mg ml⁻¹ in tetrahydrofuran was spin-coated on cleaned ITO substrates and annealed at 100 °C for 10 min. The as-prepared perovskite precursor solution was spin-coated onto the modified ITO substrate at speeds of 1,000 rpm for 5 s and 4,000 rpm for 30 s. During the last 15 s of the spinning process, the liquid film was treated by drop-casting of chlorobenzene solvent (150 µl). For the target device the doped solution was spin-coated directly on cleaned ITO substrates. Substrates were annealed on a hotplate at 100 °C for 50 min, then PEABr (1 mg ml⁻¹ in isopropanol and DMSO mixed solvent, volume ratio 200:1) were dynamically spin-coated on top (4,000 rpm, 30 s) and annealed at 100 °C for 5 min. A layer of PC₆₁BM (20 mg ml⁻¹ in chlorobenzene, 1,500 rpm, 35 s) was coated and annealed at 100 °C for 10 min, followed by spin-coating with BCP at a concentration of 0.5 mg ml⁻¹ in isopropanol on top (4,000 rpm, 30 s). Finally, Ag (100 nm thickness) was deposited on top by thermal evaporation.

Preparation of pristine and molecule-doped perovskite precursor solutions

A mixture of 6.7 mg of MABr, 8.8 mg of MACl, 18.2 mg of CsI, 196.1 mg of FAI, 22 mg of PbBr₂ and 572 mg of PbI₂ was dissolved in 865 µl of DMF/DMSO solution (volume ratio 4:1) and stirred at 60 °C for 1 h into the prepared undoped precursor solution. For the doped solution, around 1–7 mg DMAcPA was added to this solution.

Materials characterization

NMR spectra were recorded on Bruker Avance 500 and Bruker Avance 400 spectrometers in CDCl₃, DMSO-d₆, CD₂Cl₂ and DMF-d₇. High-resolution mass spectra were recorded on a Thermo Scientific Q Exactive mass spectrometer using electrospray ionization as ionization mode. Thermal gravimetric analysis was recorded on a Mettler TGA1 at a heating rate of 10 °C min⁻¹ under nitrogen flow. Ultraviolet-visible spectra were recorded on a PerkinElmer Lambda 950 spectrometer. Steady-state and time-resolved photoluminescence spectra were measured using a spectrofluorometer (FS5, Edinburgh instruments), with a 405 nm pulsed laser used as excitation source for measurements.

Device characterization

J–*V* measurements were carried out using a Keithley 2400 source meter in an ambient environment (roughly 20 °C and 60% relative humidity). The unencapsulated devices were measured both in reverse scan (1.2 to –0.2 V, steps of 0.01 V) and forward scan (–0.2 to 1.2 V, steps of 0.01 V) with 200 ms delay time. Illumination was provided by an Oriel Sol3A solar simulator with an AM 1.5G spectrum, and light intensity of 100 mW cm⁻² was calibrated by a standard KG-5 Si diode. The active area (0.08 cm²) of our device was calibrated with a shadow mask during measurements. Electrochemical impedance spectroscopy and Mott–Schottky plots were conducted on the Zahner Zennium electrochemical workstation. EQE measurements for devices were carried out with an Enli-Tech EQE measurement system, and light intensity at each wavelength was

calibrated with a standard single-crystal Si photovoltaic cell. For stabilized power output tests under illumination, bias voltage was fixed at the maximum power point voltage (*V*_{max}) and current density variation in an ambient environment was recorded without controlling the device temperature. Long-term operational stability was conducted with the encapsulated devices under a 1-sun-equivalent light-emitting diode lamp under an ambient environment (approximately 25 °C and 50% relative humidity). The maximum power point was recorded every 60 s.

Film characterization

X-ray diffraction spectra were measured using a Bruker D8 ADVANCE. XPS measurements were carried out on an Omicron ESCA Probe XPS spectrometer (Thermo Scientific ESCALAB 250Xi) using 150 eV pass energy and 1 eV step size for survey scan and 20 eV pass energy and 0.01 eV step size for fine scan. Contact angle measurements were carried out with a KLA-Tencore D120 surface profiler. SEM was characterized using a HITACHI SU8230, and FTIR by a Bruker Vertex 70v.

AFM-IR

The spatial distribution of dopants was characterized with the nanoIR2 system (Anasys Instruments). Both height and chemical images were obtained in contact mode using a Contact Mode AFM Probe for nanoIR2 in an ambient atmosphere. The chemical image was recorded at 1,410 cm⁻¹ with the specific stretching vibration of P–C in DMAcPA.

ToF-SIMS

Depth profiling data were obtained with the ToF-SIMS 5 system from ION-TOF. Bi³⁺ ions were used as primary ions, and negative ions were also detected. Sputtering was performed using Cs⁺ sputtering ions with 500 eV ion energy, 50 nA ion current and 200 × 200 µm² raster size. An area of 50 × 50 µm² was analysed using Bi³⁺ ions with 30 keV ion energy.

UPS

To measure work function, UPS spectra were recorded using an imaging photoelectron spectrometer (Axis Ultra, Kratos Analytical) with a non-monochromated He Iα photon source (*h*_ν = 21.22 eV). Au was taken as a reference.

DLTS

DLTS measurements were performed with a Phystech FT-1230 HERA-DLTS system. A modified Boonton 7200 capacitance meter (1–75 kHz) was used to examine dynamic capacitance. The effective test area was about 0.14 cm². Samples were then placed in a liquid-helium cryostat (Lakeshore 335,336). The DLTS temperature scan range was 370–205 K at 2 K intervals.

Transient absorption spectroscopy

Femtosecond transient absorption spectra were characterized by a setup based on a regenerative amplified Ti/sapphire laser system from Coherent (800 nm, 35 fs, 6 mJ per pulse and 1 kHz repetition rate), non-linear frequency-mixing techniques and a Femto-TA100 spectrometer (Time-Tech Spectra).

Measurement of pH

The pH of the precursor solution was measured using a Rex Electric Chemical E-301-QC pH meter, which was calibrated with three standard solutions before each test. After calibration the meter was washed several times with pure solvent then each solution was tested.

Hall measurement

Hall measurement was performed on a Nanometrics HL5500 Hall system at room temperature and under darkness. Samples were prepared in Van der Pauw geometry. Connections of silver wires for electrical measurement were made using conductive silver paste. After drying the paste, the ohmic behaviour of metal contacts was S4 checked by

the linear variation of current–voltage characteristics, which were observed to be independent of the polarity of the applied current and contact combinations for each sample. Measurements were done under a 0.5 T magnetic field; sample size was $15 \times 15 \text{ mm}^2$.

DFT calculation

All first-principles calculations were carried out based on DFT, as implemented in the Vienna Ab initio Simulation Package²⁶. The Perdew–Burke–Ernzerhof form of exchange–correlation functional²⁵ was utilized. Plane-wave energy cutoff was 290 eV, together with projector-augmented wave pseudopotentials. A single gamma point was used for *k*-point sampling for molecular systems and a $1 \times 2 \times 1$ *k*-point mesh for slabs. Structures were optimized with an energy and force tolerance of 10^{-6} eV and $0.02 \text{ eV } \text{\AA}^{-1}$, respectively. Surface adsorption was carried out on a PbI_2 -terminated FAPbI_3 slab of size $19.2 \times 12.8 \times 11.3 \text{ \AA}^3$, together with a vacuum layer of $21 \text{ \AA}$ in the out-of-plane direction. Binding energy (E_b) is defined by $E_b = E(\text{complex}) - \sum E(\text{component})$. Van der Waals interactions in dimers were simulated using the DFT-D3 approach³⁷. All DFT calculations were carried out on the Taiyi cluster supported by the Center for Computational Science and Engineering of Southern University of Science and Technology.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data are available in either the main text or supplementary materials, and are also available from the corresponding author on reasonable request.

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Author contributions Z.H., Q.T. and Z.L. conceived the idea for the project. Z.H. directed and supervised the project. Q.T. fabricated and characterized the devices. Z.L. designed and synthesized molecules and carried out NMR and FTIR characterization. G.L. performed DFT calculations. X.Z. conducted AFM–IR characterization. B.C. and T.C. carried out DLTS measurements and data analysis. G.C. and H.Y. performed UPS characterization. H.G. conducted SEM and energy-dispersive X-ray spectroscopy. G.M. conducted Hall testing. D.H. performed pH and other electrochemical testing. J.W. and J.X. participated in characterization of other properties. Z.H., Z.L. and Q.T. analysed all data and wrote the manuscript. All authors discussed the results and commented on the manuscript.

Competing interests A Chinese patent application (no. 2023100349666), submitted by SUSTech, covers the AcPA series molecule-based molecule-extrusion process for the fabrication of HTL-free perovskite solar cells.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-023-06207-0>.

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► Experimental design

Please check: are the following details reported in the manuscript?

1. Dimensions

Area of the tested solar cells	☒ Yes ☐ No	The area of tested solar cells is 8.02 mm square
Method used to determine the device area	☒ Yes ☐ No	Shadow mask

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction	☒ Yes ☐ No	We provide forward and backward scans in Figure 5A and certificate.
Voltage scan conditions <i>For instance: scan direction, speed, dwell times</i>	☒ Yes ☐ No	Time intervals of 100ms at each measured voltage with step size of 10 mV were employed for the measurements. Thus, the scan speed is approximately 100 mV/s. Both forward and reverse scan speeds are identical. See more details in main text.
Test environment <i>For instance: characterization temperature, in air or in glove box</i>	☒ Yes ☐ No	The atmosphere temperature during J-V test is 23-25 degree C , and the temperature of the tested cell was unknown. We measured the J-V scan in air with related humidity of 50-60 %.
Protocol for preconditioning of the device before its characterization	☐ Yes ☒ No	No preconditioning protocol was used.
Stability of the J-V characteristic <i>Verified with time evolution of the maximum power point or with the photocurrent at maximum power point; see ref. 7 for details.</i>	☒ Yes ☐ No	We provide 500s maximum power point tracking data in Figure S29 to evaluate the stabilized PCE.

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization	☐ Yes ☒ No	No unusual behavior and negligible hysteresis were observed.
Related experimental data	☒ Yes ☐ No	We provide J-V hysteresis scans in Figure 5A and certificate.

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)	☒ Yes ☐ No	We provide EQE measurement in Figure 5B.
A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes ☐ No	We compared the integrated Jsc with the one from JV scan. The difference between the integrated Jsc from EQE and the Jsc from the JV scan is within 4% difference which is within the accuracy confidence of the measurements. See more details in the main text.
For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐ Yes ☒ No	Explain why this information is not reported/not relevant.

5. Calibration

Light source and reference cell or sensor used for the characterization

☒ Yes
☐ No

Illumination using an Oriel Sol3A solar simulator with AM1.5G spectrum and light intensity of approximately 100 mW/cm² was calibrated by a standard KG-5 Si diode. The mismatch factor was calculated to be less than 2%, and this mismatch correction was applied to achieve the equivalent of 100 mW/cm² AM1.5 irradiance for the 1 sun measurements. See details in device characteristics in the methods section.

Confirmation that the reference cell was calibrated and certified

☒ Yes
☐ No

Our solar simulator was calibrated by KG-5 Si solar cells (certificated by Newport).

Calculation of spectral mismatch between the reference cell and the devices under test

☒ Yes
☐ No

A spectral mismatch calculation was performed based on the spectral intensity of the solar simulator and 3 typical EQEs of our cells. This resulted in 2 mismatch factors of $M = 0.9951$ and 0.9968 .

6. Mask/aperture

Size of the mask/aperture used during testing

☒ Yes
☐ No

An optical aperture mask (8.02 mm square) was used for measuring our cells.

Variation of the measured short-circuit current density with the mask/aperture area

☒ Yes
☐ No

No significant variations were observed.

7. Performance certification

Identity of the independent certification laboratory that confirmed the photovoltaic performance

☒ Yes
☐ No

Certification was performed in an external independent institute-Shanghai Institute of Microsystem and Information Technology (SIMIT) in Shanghai, P.R China.

A copy of any certificate(s)
Provide in Supplementary Information

☒ Yes
☐ No

We provided a soft copy of certificate in Figure S31.

8. Statistics

Number of solar cells tested

☒ Yes
☐ No

Overall we fabricated and tested more than 500 cells for this study. In the statistics, the results of 24 devices were used for each kind of devices.

Statistical analysis of the device performance

☒ Yes
☐ No

Statistical analysis were showed in Figure S28, 30, 33, 35.

9. Long-term stability analysis

Type of analysis, bias conditions and environmental conditions

☒ Yes
☐ No

Device stabilities under light conditions were performed and the results were shown in Figure 5C. See more details in methods section of Supporting Information.

For instance: illumination type, temperature, atmosphere humidity, encapsulation method, preconditioning temperature